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Praxis Precision Medicines Inc. (PRAX): Mined the Gap – How PRAX Twisted Dropouts into a Phase 3 "Win" with Make-Believe Data and Half-Baked Analysis

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"You're a fraud and you know it / But it's too good to throw it all away
You've got 'em going / And you're careful not to show it / Sometimes you even fool yourself a bit / It's like magic / But it's always been a smoke and mirrors game…

So now that you've arrived, well you wonder / What is it that you've done to make the grade?"

– Gotye, Smoke and Mirrors

We are short Praxis Precision Medicines Inc. ("PRAX", "the Company"), which in October 2025 announced "positive" results from its Phase 3 trial of ulixacaltamide ("ulixa", "the drug") to treat essential tremor ("ET"). We believe the readout has been effectively engineered by Praxis through a combination of (a) a last-minute endpoint change – which has never received FDA blessing – in order to inflate treatment effect, (b) imputing over-optimistic results for the astronomical 36% of treatment arm patients who discontinued, then (c) feigning "conservatism" in its final analysis by blatantly misapplying the very methodology described in the paper the Company cites as support. **In our view, any single one of these points could individually doom FDA approval hopes, but taken together, they doom ulixa's chances entirely.** Our views are informed by our consultations with experts, including former Praxis directors, a biostatistician, and four former FDA officials who each spent between 10 and 40 years at the agency. As Praxis approaches its Q4 2025 pre-NDA meeting, we believe shares hold substantial downside as the FDA sees through the Company's smoke and mirrors game.

The vast majority of Praxis' valuation is now tied up in ulixa, which the Company acquired in 2018 for a mere $1 million, a telling sign of its lack of promise after its prior owners failed to develop the drug. Further, three other drugs with ulixa's mechanism have already failed early stage ET studies. For its part, ulixa *also* failed its Phase 2 trial ("Essential1"), but Praxis continued onto Phase 3 ("Essential3") regardless. Then in March 2025, Essential3's independent data monitoring committee ("IDMC") recommended the Phase 3 trial be halted for futility. Praxis again pushed on, we believe, because the Company found multiple ways to game its way into a positive readout.

Indeed, immediately after reporting "positive" Phase 3 topline results, Praxis raised $567 million in capital. Sell-side analyst notes now cheer on ulixa's potential; notes we reviewed suggest upwards of 80% approval odds and a $3 billion valuation for ulixa relative to the Company's $4.5 billion market cap. We think ulixa is a house of cards.

Last-Second Primary Endpoint Change, Unapproved by the FDA

In February 2025, Essential3's IDMC recommended the trial be halted for futility, even as CEO Souza claimed this was a sub-3% probability just weeks earlier. Praxis pushed ahead, citing the Company's ability to explore "*different analysis methods for the final dataset.*"

In September 2025, Praxis abruptly changed the Phase 3 primary endpoint from Day 84 to Day 56. CEO Souza claims Praxis conducted "extensive simulations" and merely "referred back to what we knew from the Essential1 program." Yet Souza's post-hoc explanation is illogical – the Company already had the Essential1 results in hand when selecting the Day 84 endpoint. Rather, we believe Praxis likely saw – even under blinded conditions amid the IDMC's halt recommendation – that discontinuations were heavily front-loaded, as 81% of pooled dropouts occurred by Day 56. As such, Praxis' decision to move to Day 56 locked in maximal statistical modeling benefits while avoiding the added variability that might bury an already questionable treatment effect.

Crucially, Praxis – despite repeated questioning – has never confirmed that the FDA approved the change. Instead, Souza says *"it doesn't matter"*, as the trial "succeeded" at both Day 56 and Day 84. Yet as we show herein, we think that ulixacaltamide would be dead in the water had it held to Day 84. Ulixa's efficacy already shows a much weaker signal at Day 84 than Day 56 – treatment effect drops by a full point – and we think likely fails in a sensitivity analysis. According to a 10-year biostatistician we consulted, *"I haven't seen them do the sensitivity analysis for Day 84. I've done some modeling and there's a good chance it doesn't hit the sensitivity analysis."*

Massive, Unbalanced Dropouts Paves Way for Aggressive Phantom Data Imputation

Phase 3 topline data reveals 35.6% dropout in the treatment group – a massive imbalance vs. 5.6% in placebo, and also far higher than the 17% treatment arm dropout seen in Phase 2. Praxis claims the drug is "well tolerated" yet the majority of dropouts cited AEs. This alone is enough to jeopardize the validity of the entire trial, according to multiple former FDA officials we spoke with: *"Generally at 25% dropout, the FDA will question validity… If more than 25% drop out, it's just not effective… The failed patients already dropped out, so how can the trial be considered successful?"*

Yet problems also lie in how Praxis dealt with the problem of these missing patients. The Company employed a mixed model for repeated measures ("MMRM") under a missing at random assumption ("MAR"). This method not only relies on balanced dropout between arms – blatantly not the case here – but aggressively assumes that patients who dropped from the study would have performed in line with patients who remained in the study. It also inflates results from front-loaded dropouts, which is also the case here. In short, Praxis imputes make-believe data showing phantom improvements in patients who quit the study days, weeks, or months earlier.

Multiple former FDA officials and a biostatistician we spoke with said this approach has zero chance of being accepted by the FDA; *"Because it's 35% versus 5%, it's clearly due to adverse events… MMRM under MAR will definitely not be accepted"*; *"there's 100% no chance the FDA accepts that";* it *"artificially suggests that a dropout does well"* especially as ulixa's treatment effect regresses past Day 56. As one put it:

> *"That part is merely for the public, not for the FDA. It definitely shouldn't be applied. They can make it public – as a public company – they try to do the most beneficial thing for them to say it's a positive trial, but there's 100% no chance the FDA accepts that."*

Experts say the FDA will re-run the missing data through a more conservative analysis, endangering results. When we utilize more conservative assumptions such as placebo-based imputation, we estimate ulixa fails its own 3.0-point benchmark at the original Day 84 endpoint.

The Tipping Point: Praxis Relies on Half Baked Analysis to Feign "Conservatism"

Praxis then argues that its aggressive if not nonsensical imputation is justified via a "tipping point" analysis that penalizes these patients by 2.5 points, equivalent to one-half standard deviation of the data. Praxis claims that this 2.5-point penalty is "the maximum" and an "extreme scenario." This is again nonsense. Praxis relies on a 2013 paper by Ratitch et al., which calls for one-half standard deviation as a *minimum* adjustment, not a *maximum* one as Souza claims. Tellingly, when we estimate Phase 3 results using the actual prescribed methodology at 0.75x and 1.0x standard deviation, ulixa fails.

Moreover, *Ratitch* cites studies with minimal dropout rates, nothing close to the 35.6% here. According to the 10-year biostatistician we consulted, *"the paper that they're citing isn't particularly relevant, and there's nothing natural about the half standard deviation."* The FDA itself says studies with higher dropout rates ought to be accompanied by a more conservative tipping point analysis, and has on multiple occasions denied approvals citing insufficient tipping point analyses, even when the sponsor promoted a "positive" readout. Key FDA officials including newly appointed CDER Director Pazdur and CBER Director Prasad have a demonstrated history of

calling out insufficient tipping point analyses, including from Amgen and Merck. We think they will have no problem rebuking Praxis' nonsense here, too.

Conclusion: We Recognize the Need for Safe and Effective ET Drugs – Ulixa Isn't It

We recognize that ET is a serious and undertreated condition that demands safe and effective therapies. That is exactly the problem, we believe, for ulixacaltamide – an ineffective drug with tolerability issues dressed up with make believe data and statistical games. The drug has already failed in Phase 2 and was recommended to be halted for futility in Phase 3. Now, its miraculous "success" rests on a last-minute endpoint change, aggressive imputation of discontinued patient data amid an alarming, highly imbalanced dropout rate; and a blatantly misapplied tipping point analysis. Each of these fall apart under scrutiny. Taken together, they damn ulixa's FDA hopes entirely.

History shows the FDA will not hesitate to reject weak data, as UniQure's experience last month makes clear. We believe Praxis' situation is even more precarious, given the combination of factors above and that the Company never appears to have received FDA blessing of its primary endpoint change. We think Praxis shares face substantial near-term downside as the Company heads into its Q4 pre-NDA meeting that is likely to trigger an FDA reckoning.

"The goal for them was to ring the bell at NASDAQ, and that concern outweighed every scientific and medical consideration… A company operating at the edge of what is professional." – Former Praxis Director

"Study 1 has big problems… they moved the primary endpoint from week 12 to 8, and this is a big ticket item in the FDA's view… In general, the FDA will not accept such modifications."
– Former 20+ Year FDA Biostatistician and Neurology Specialist

"They didn't get approval from the FDA to change the endpoints…"
– 10-year Biostatistician at a Top University

"Generally at 25% [dropout rate] the FDA will ask about the validity of the trial… The failed patients have already dropped out so how can you do a valid trial or consider that successful? …It's hard to see how there's efficacy [if] you can't take the drug. It's a major problem…"
– Former 20+ Year FDA Biostatistician focused in Neurology

"We're not talking about 3% or 5% of missing data, we're talking about a hell of a lot of missing data… The FDA will present this quandary to this sponsor, and they won't suffer fools easily… Houston, we have a problem! Would I buy the stock? I don't think so. There's too much damn missing data." – Former 40-Year FDA Pharmacology/Toxicology Specialist

"When the company does really low sensitivity to try to convince the FDA. To me that's a specific concern… that analysis will definitely not be accepted by the FDA … As a public company, they may try to do the most beneficial thing for them to say it's a positive trial, but there's 100% no chance the FDA accepts that."
– Former 20-Year FDA Medical Officer and Neurology Specialist

"It's not a fair presentation of the results. Adjusting people by 2.5 points is imputing them as if the treatment is conferring benefit… I'm not convinced by that approach. And then they're not telling us what the actual tipping point is. I mean, come on… A new dataset will be needed."
– 10-year Biostatistician at a Top University

Introduction to Praxis, ulixacaltamide, and Essential Tremor

Praxis was founded in 2015 as EpiPM Therapeutics, and now refers to itself as *"a clinical-stage biopharmaceutical company translating genetic insights into the development of therapies for central nervous system (CNS) disorders characterized by neuronal excitation-inhibition imbalance."*

The Company's flagship drug candidate is ulixacaltamide ("ulixa"; "PRAX-944"). In January 2018, Praxis acquired ulix ("PRAX-944") from EPIRUS Biopharmaceuticals, Inc (EPRS.Q) – which had spent several years studying the compound for potential use in neuropathic pain – for a mere $1 million. We find the paltry price tag indicative of its poor clinical prospects.

In October 2020, Praxis completed a $190 million IPO on the NASDAQ GS, touting PRAX-944 as one of its leading drug candidates for potential use in treating essential tremor ("ET"). ET is a nervous system disorder affecting roughly 7 million people in the U.S., primarily those above 65 years of age. Its primary symptoms are trembling in the hands, such that the disorder is often confused with Parkinson's. ET is thought to be caused by a gene alteration passed from parent to child. Current first line medications include beta blockers such as propranolol and anti-seizure medications such as primidone and gabapentin. According to Praxis, up to 50% of ET patients don't receive treatment altogether and 77% of ET patients don't feel their symptoms are manageable with current treatments.

Ulixa is a T-type calcium channel inhibitor, which the Company argues is linked to the abnormal brain activity that manifests with ET. While we acknowledge that neurology is incredibly complex, we have seen little evidence to suggest that this mechanism can succeed in ET. On the contrary, **three drugs with the same mechanism have already been tested and failed earlier stage clinical trials in ET.**

- Jazz Pharmaceuticals plc (JAZZ) studied suvecaltamide (JZP385), a highly selective and state-dependent modulator of T-type calcium channels, in adult patients with ET. Suvecaltamide did not achieve statistical significance at 30mg versus placebo on the primary endpoint or the key secondary endpoint. One participant experienced a serious adverse event considered treatment related by the investigator.

- Neurocrine Biosciences studied apinocaltamide (NBI-827104), a triple-T calcium channel blocker. The drug failed to meet primary or secondary efficacy endpoints in Neurocrine's Phase 2 study.

- Ethosuximide, another T-type calcium channel inhibitor, failed to show that it was an effective treatment. An open label trial initially planned to enroll 15 patients, but stopped the trial after 7 patients participated due to *"lack of efficacy and a high incidence of adverse effects".*

Tellingly, Praxis' earlier trials of ulixa have shown mixed results, at best. In March 2023, Praxis reported negative top-line results from a Phase 2b dose-range-finding trial of ulixa for ET ("Essential1"). The study measured the change in TETRAS modified Activities of Daily Living ("mADL") in 132 patients over 56 days. Ulixacaltamide failed to produce a statistically significant difference to placebo (p=0.126). Shares fell over 40% overnight.[1]

[1] Note the 17% discontinuation in the treatment arm is far lower than what Praxis later reported in both Study 1 and Study 2 of Phase 3.

Praxis Phase 2b	Ulixacaltamide	Placebo
Patients	78	38
Discontinuations	13	4
Rate	17.0%	11.0%
Due to AEs	9	1
Rate	11.5%	2.6%
mITT population	78	38
mADL (primary endpoint)	(3.01)	(1.44)
p value	**0.126**	

Moreover, while investors often like to see dose-dependency – i.e., a correlation between dosage and therapeutic effect, which might support a positive view of the drug's mechanism – the Phase 2b showed no dose-dependency.

Nevertheless, CEO Souza argued that ulixa produced positive results for certain secondary endpoints, and that the Company would proceed to a Phase 3 trial ("Essential3"). Essential3 is composed of two studies, each of which measures endpoints tied to "modified activities of daily living 11 ("mADL11") scores.[2]

- Study 1 is placebo-controlled parallel group study over 12 weeks, (n=199 drug vs. n=233 placebo). The primary endpoint measures differences in responses of mADL11 scores. **Our views pertain primarily to this study.**

- Study 2 is randomized withdrawal study, which took 80 responders (i.e., those who showed an improvement of 3+ mADL11 points), and after 8 weeks, split responders into two equal groups of 40. One group then received the drug, while the other group reverted back to placebo. The primary endpoint measured the percentage of each group that continued to show a clinical response after an additional 4 weeks.

On October 16, 2025, PRAX reported "positive" top line results from each of these studies. Shares rallied 185%, and the Company promptly raised $567 million in an equity offering. We note that the Company's proxy statement discloses executive bonuses are based on the achievement of *"our corporate objectives for such year"* and that *"the metrics we use for incentive awards"* include, *"clinical, research and development"* objectives. Last year, CEO Souza earned total compensation of $13.4 million.

How Praxis Reverse Engineered a Phase 3 Readout: Shifted Endpoints, Phantom Data Imputation, and a Blatantly Mishandled Tipping Point Methodology

We believe Praxis effectively engineered a Phase 3 "win" through shifting endpoints, highly aggressive missing data imputation, and a blatantly misapplied tipping point analysis.

- We believe Praxis would not have achieved statistical significance with a proper tipping point analysis at the Day 84 endpoint. The fact that Praxis discloses only a "select sensitivity analysis" for Day 84 mADL11 described as *"the average of Day 77 and Day 84 at the time of the interim analysis"* is indicative. We cannot find any prior reference in Praxis materials to an endpoint defined as an average of Day 77 and Day 84 – it appears entirely contrived to support the Company's preferred narrative.

[2] These are patient-reported scores, obtained through an 11-question survey that assesses the extent to which tremors limit patients' ability to complete everyday tasks. Scores are ranked on a scale of 0 to 33, where 33 is the worst and 0 is the best.

- We estimate that with more conservative imputation assumptions (i.e., placebo-based), ulixa efficacy falls from a 4.3 point change in mADL11 to just 3.79 points at Day 56, and then at Day 84, falls to just 2.91 points – well below the Company's own definition of responders at 3.0 point minimum change.

- We finally estimate that – even at its preferred Day 56 endpoint – had the Company conducted a proper tipping point analysis which *begins* rather than *ends* at 0.5 standard deviation penalties, ulixa's perceived efficacy would once again collapse. At 0.75 standard deviations, ulixa shows a 2.8 point change in mADL11 (again – well below the Company's own 3 point minimum). At 1 standard deviation, not only does ulixa show just a 2.3 point change, but this represents a mere 0.8 point difference vs. placebo, rendering the p-value = 0.156 – a horrific failure.

Individually, any one of these elements exposes ulixa's claimed efficacy. Together, they expose how the results have been generated by aggression, manipulation, and misrepresentation rather than actual performance.

Praxis Misrepresents Motivations For its Last-Second Primary Endpoint Change, Which We Believe Lacks FDA Blessing

In February 2025, Praxis disclosed that the Phase 3 independent data monitoring committee ("IDMC") recommended that the study be stopped for futility, *"due to the results being unlikely to meet the primary efficacy endpoint under the parameters set by the statistical model."* PRAX shares fell 47% on the news.

Just a month earlier, Praxis CEO Souza had suggested that this outcome was nearly impossible – a *"less than 3% probability of being futile, the way it was designed."*

Yet Praxis carried on, with CEO Souza stating that, *"following the advice of the committee, we will explore different analysis methods for the final dataset, which is expected in the third quarter of 2025."*

In September 2025, Praxis changed the trial endpoints from change in mADL11 at Day 84 (12 weeks) to Day 56 (8 weeks). The change is not only a red flag in light of the IDMC halt recommendation, but as CEO Souza provided an entirely nonsensical explanation of the change.

> *"Between that point and last month, we conducted extensive simulations and analysis on the blinded data. We consulted with a number of statistical, clinical and regulatory advisers, and we decided that the most prudent thing to do was to refer back to what we knew from the Essential1 program that was the ability for ulixacaltamide to positively modify the disease when patients are exposed between baseline and day 56. So we made the decision to change the primary endpoint to day 56 from day 84."*

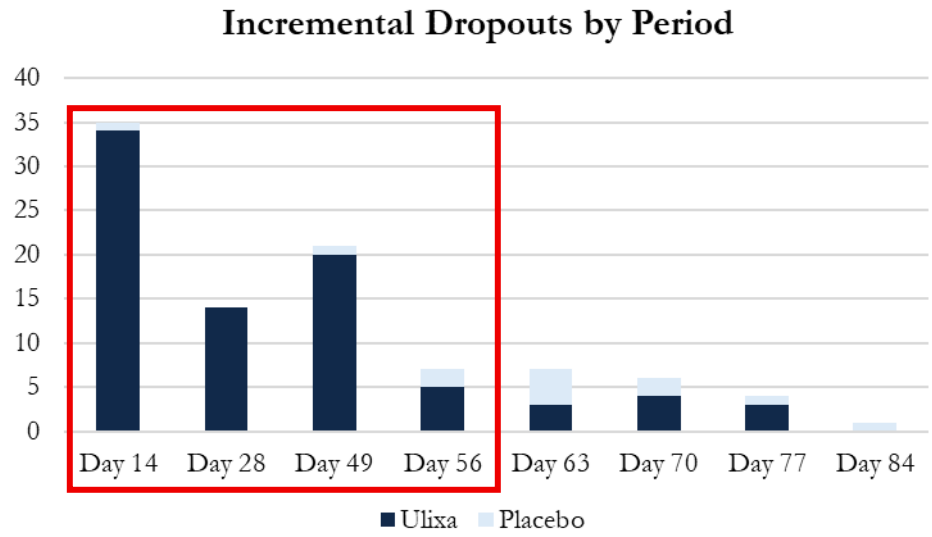
CEO Souza suggests that the Company decided to simply refer back to Essential1, but this doesn't stand the common sense test. Praxis was *already* aware of ulixacaltamide's performance in Essential1 prior to picking the Essential3 day 84 primary endpoint. According to one biostatistician we consulted,

> "*There are other problems with the design of the trial that make me uncomfortable. I would not be comfortable advising a sponsor to do these. One of those is changing the primary endpoint. Day 56 to 84 then back to Day 56. I really don't know why they would do that. "*

In the words of a second expert – a former 10+ year FDA Neurology official we spoke with,

> *"For some reason they decided to use 56. The rationale they published is that the phase 2b was positive. I'm not really sure the rationale there is accepted by the FDA. I suppose they thought the power was sufficient at day 56. If that was the case, then why wouldn't you find it at day 84?"*

We believe Praxis is misrepresenting its true motivations. CEO Souza's admission that the Company ran *"extensive simulations"* on interim Phase 3 data is suggestive. Consider that Praxis likely saw – evident even in pooled blinded data – that the study suffered from a massive number of front-loaded dropouts. Inclusive of the ITT population (patients who dropped prior to a post-baseline reading, which would have been included in interim data), 78 of 96 (81%) total dropouts occurred by Day 56.



Source: Praxis Phase 3 Data Presentation, slide 14

As such, by Day 56, the core bias from missing at random ("MAR") imputation – which treats early dropouts as if they perform in-line with completed patients – would *already* be essentially maximized. Thus, the Day 84 endpoint adds extremely little favorable MAR imputation bias, but *does add meaningful placebo variance*, hence diluting an already modest efficacy signal. In short, Praxis – even with blinded data – possessed both the means and the motive to alter the primary efficacy endpoint in the Company's favor.

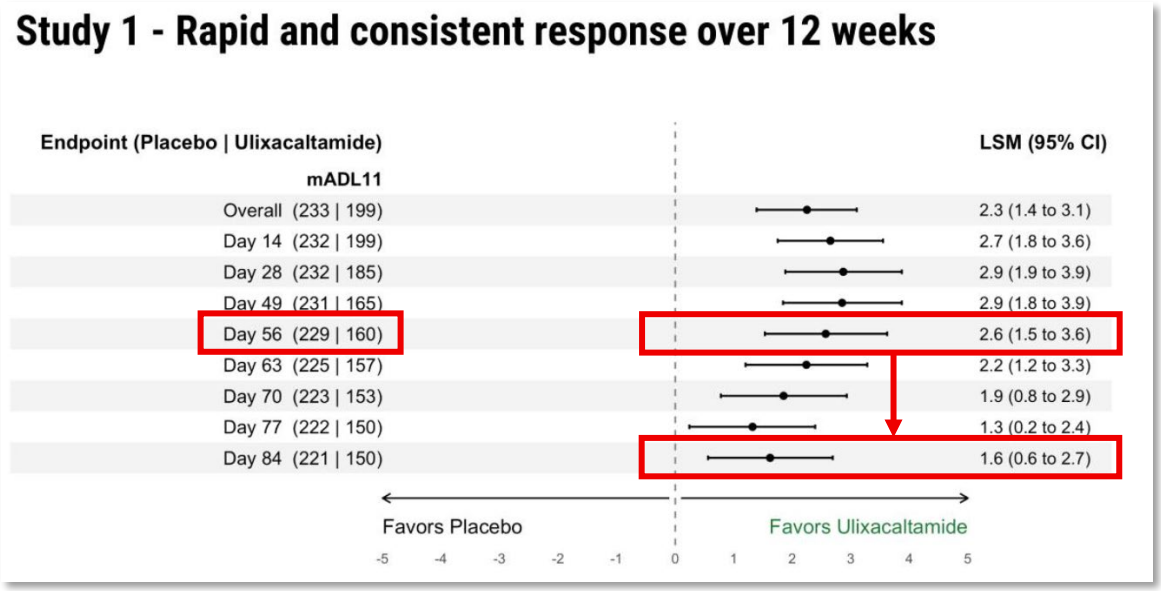
To our knowledge, Praxis has never explicitly told investors on calls or in filings that the FDA has signed off on this primary endpoint change. Instead, CEO Souza has deployed squishy language to defend the Company against the obvious implication of foul play.

- On the Company's October 16, 2025, special call, Souza took pains to emphasize that Praxis, *"consulted with a number of statistical, clinical and regulatory advisers"* when it *"simplified the key secondary endpoints based on prior feedback from the FDA"* and that *"**we're seeking** and gained alignment on the Phase III program as it relates to the endpoints, to dose and the overall design."* Yet this very language implies that Praxis *did not in fact* receive explicit FDA blessing of the change.

- Just two days ago at the Jefferies London conference, Souza again evaded questioning as to FDA blessing, instead pivoting to suggest that the endpoint change simply doesn't matter, as the trial was successful at either endpoint.

  "The choice of the time of the assessment is always the sponsor's. We know that it's acceptable to do 8 weeks or 12 weeks. We did change, I want to be incredibly clear about that from 12 weeks as assessed at the average of day 77 and day 84 to day 56, so pretty minor change but maybe even more important than that is the study was successful at every timepoint, including the one that it was changed from."

We think otherwise. Praxis' own data – despite the Company's claims that ulixa has a *"consistent response over 12 weeks"* – shows that treatment effect waned by a full point from Day 56 to Day 84.



Source: Praxis Investor Presentation

According to a former 15-year FDA Neurology official we spoke with,

> *"You could ask the question 'is 1.6 good enough?' and typically we think of a 2.0 point [difference] as good enough. No matter what you do, it's [treatment effect] less than 2 when you go out past 60 days.*[3] *It's at risk on the clinical meaningfulness… The degradation of response is a real risk to it [FDA acceptance]."*

Phase 3 Trial Plagued by High and Unbalanced 36% Treatment Arm Dropout Rate; Praxis Aggressively Imputes Improvements to a Massive Number of Missing Patients

Praxis claims that ulixa was "well tolerated," but 35.6% of patients discontinued treatment, with adverse events accounting for most of those cases – 27% of all patients. This presents a red flag both as this dropout rate is far higher than seen in Phase 2 trials and at odds with Praxis' claims that the drug is "well tolerated." Even in isolation, this staggering dropout rate could derail the Company's FDA hopes, according to multiple experts we spoke with.

> *"Generally at 25% [dropout rate], the FDA will ask about the validity of the trial. If more than 25% drop out, it's just not effective… The drug had some AEs, but they weren't very severe, so the FDA will ask them to explain that. Because the Company claimed it was tolerated. But in the real world, compliance will be even worse…* ***The failed patients already dropped out, so how can the trial be considered successful?*** *That's the major problem… The efficacy is an issue; the compliance is an issue."*
> – Former 20-year FDA Medical Officer focused in Neurology

Yet our concern is not solely with the sheer number of patient dropouts, but with how Praxis accounts for these patients: **by aggressively imputing phantom improvements for nonexistent patients, which tips the scales in the Company's favor.** Clinical trial sponsors can handle the issue of dropouts in multiple ways.

[3] To be clear, the data shows treatment effect of 2.2 points at Day 63 and 1.9 points at Day 70.

- Last observation carried forward fills missing values with the most recently recorded value.
- Worst rank fills missing values with the worst recorded value (conservative if not overly penalizing).
- Placebo-based imputation fills missing values with values in-line with placebo outcomes.

Yet Praxis employed a mixed model for repeated measures ("MMRM") under a missing-at-random assumption ("MAR"). This imputation method, per the Company, treats the *"subject as a random effect with an unstructured covariance matrix."* In layman's terms, the model fills missing values **with data that assumes patients who dropped from the study would have performed similarly to those who remained.**[4] In our view, this is a wildly aggressive, unsuitable methodology that artificially inflates results.

Numerous former FDA officials we spoke with suggested that this methodology was wholly unsuitable, especially given the massive disparity in dropout rates between the groups.[5] Per one former 10+ year FDA official,

> *"When I read the Company's claim, they claim they use MMRM, and* ***that analysis will definitely not be accepted by the FDA.*** *The definition is random – it assumes both treatment and placebo drop out [at equivalent rates]. And if you claim you're getting a positive result with that methodology, they [Praxis] won't get the FDA [approval]. Because it's 35% [treatment dropout] vs. 5% [placebo dropout], it's clearly due to the adverse events in the treatment group.* ***It's clearly not correct to use that****… You need to be missing randomly. Missingness is much higher in treatment group than placebo, so the high dropout is not missing randomly."*

We asked another 20-year FDA Biostatistician what they would have recommended Praxis do in this situation:

> *"You should assume the dropout patients are performing worse, in line with the placebo data. From that time on, I'd impute the placebo value for the patients."*

We also spoke with a 15+ year FDA official who emphasized that the Company's methodology creates an artificially favorable result, especially as Praxis' own data shows that treatment effects *decline* past Day 56.

> *"Because say they drop out at Day 49, and they were doing well at that time. It [imputation] has the inherent problem that it* ***artificially suggests that a dropout does well****. That's the kind of concern that the FDA will be all over, because if it regresses after three months to baseline, then does it have a sustained effect?"*

They noted that alternative handling of missing patient data would lead to worse outcomes, but that the FDA would ask for the full dataset and run their own analysis using varying (less Company-favorable) imputation methods, which could cause significant problems for Praxis.

> *"The last observation carried forward is in this case not going to work, partly because the treatment effect declines over time."*
>
> *"The FDA will ask for the full dataset to Day 84, and run it on the final data. They will run their own analysis with all three versions of the imputation. And the risk is the FDA will look at that, and say 'we want you to have a longer study.'"*

[4] See additional commentary on MMRM and MAR here and here.
[5] Other researchers have highlighted the biases that can arise when dropouts are unbalanced between arms or not, in fact, missing at random. Consider here and here. If dropout is related to unobserved outcome (MNAR), which is often plausible in differential dropout due to efficacy or side effects, the MAR assumption will be violated and the MMRM may produce a biased result.

Another 20-year FDA veteran we spoke with said the same thing – the FDA will simply impute its own values using more conservative methods.

> *"The FDA would do two or three analyses. The most comparative would be to show if the dropouts were totally ineffective. Another way would be to use the last data recorded."*

As shown below, we estimate that – had Praxis used placebo-based imputation, the treatment effect would have fallen meaningfully from 2.6 points to just 2.1 points, while **on the Company's original Day 84 endpoint, treatment effect would have declined even further to a mere 1.2 points**.

mADL11 declines	Ulixacaltamide	Placebo	Difference
Day 56 (reported)	(4.3)	(1.7)	(2.6)
Day 84 (reported)	UNDISCLOSED	UNDISCLOSED	(1.6)
Day 84 (culper est.)	(3.3)	(1.7)	(1.6)
mITT patients	199	233	
Discontinued by Day 56	39	4	
Discontinued by Day 84	49	12	
Culper Est. Placebo Imputation			
Day 56	(3.79)	(1.7)	(2.09)
Day 84 (original endpoint)	**(2.91)**	**(1.7)**	**(1.21)**

Indeed, Praxis' ridiculous imputation games may help explain *why* treatment effects appear to decline past Day 56: Praxis benefits from imputing phantom improvements in dropped patients, and the bulk of these patients discontinued *prior to* Day 56. **Consistent with our view of ulixa's lack of efficacy – the perceived treatment effect at Day 56 was never real in the first place.**

We asked one former FDA official why Praxis would rely on such aggressive imputations, if it were so obviously incorrect to do so. Their response was telling.

> *"That part is merely for the public, not for the FDA. It definitely shouldn't be applied. They can make it public – **as a public company – they try to do the most beneficial thing for them to say it's a positive trial, but there's 100% no chance the FDA accepts that**."*

Nevertheless, Praxis argues that its trial is "robust" as it has been stress tested against a "tipping point" analysis. Here too, however, we believe the Company has again deployed another set of shenanigans – feigning a real stress test by blatantly misapplying the very methodology it claims to rely on.

Praxis' "Tipping Point" Analysis Relies on Blatant Misrepresentation of a 2013 Paper That Calls for a 2.5-Point Penalty ***at Minimum*** – Praxis Uses 2.5 as a ***Maximum***

Tipping point analysis seeks to stress tests the effect of imputed data on the results of a clinical trial, by penalizing the imputed data by a certain margin – sensitizing it in case the patients indeed performed worse than the imputed data suggests. If it takes extreme, unrealistic assumptions of the imputed data to overturn the trial success, then the trial is robust.

Praxis argues that its tipping point analysis demonstrated that – even after penalizing the imputed patients by 2.5 mADL11 points – the trial still resulted in a win. CEO Souza justifies this 2.5-point figure as *"up to what has been defined as **the maximum you should do, that is about half a standard deviation** or so…"*

Brian Peter Skorney Robert W. Baird & Co. Incorporated, Research Division – Senior Research Analyst

Congrats on the great data. Just a couple for me. Maybe I missed this, but can you walk through how discontinuations were accounted for in the mITT analysis? And have you run a tipping point analysis on the results yet?

And then my second question is, I know there are nonscheduled drugs that can show euphoria and there's precedent with type calcium channels for not being scheduled. But given the delta that you see on euphoria, can you discuss any regulatory discussion around evaluation of abuse potential?

Marcio Silva De'Souza Praxis Precision Medicines, Inc. – President, CEO & Director

Absolutely, Brian. So the -- as you know, right, by the very follow-up of your question, the [MMRM in MDD] assumes as a condition of the model that the data is missing at random or a MAR assumption. We prespecified and agreed with the agents that we would conduct a tipping point analysis. And I know the press release is very large, but if you look into the bottom of the table on the Study 1, we're already showing you the results of that tipping point analysis, right?

Tipping points are, as you know, gradual because you keep like going up around to what has been defined as the maximum you should do, that is about half a standard deviation or so in our case, it was around 2.5 points. you normally break or you tip. In our case, when you get there, the p-value was 0.0026. So we did not tip. In other words, as you know, but for the others, not so worse in the call, means it doesn't matter, right? You can really attempt to break by reproducing MAR data pattern here and the data is still incredibly strong. So very, very happy with that.

Source: October 16, 2025 Conference Call

As shown, Praxis provided a sensitivity analysis confirming *"robustness"* of the trial at a 2.5-point shift, *"exceeding the ~1/2 SD robustness criterion of Ratitch et al. (2013)."* The Company's investor presentation similarly claims that *"efficacy remains significant under extreme scenarios."*

mITT population	Ulixacaltamide (n=199)	Placebo (n=233)	p-value
Primary Endpoint			
Day 56 CFB mADL11	-4.3	-1.7	<0.0001
Key Secondary Endpoints			
Rate of Disease Improvement, Baseline to Day 56 mADL11	-4.0	-1.7	<0.0001
PGI-C Day 56	3.3	3.9	<0.0001
CGI-S CFB to Day 56	-0.41	-0.12	0.0007
Select Sensitivity Analyses			
Imputation of missing data for primary analysis*	-3.3	-1.6	0.0026
Day 84 CFB mADL11**	-3.4	-1.9	0.0049

* Results from a pre-specified delta-adjusted tipping-point analysis remained statistically significant at the maximum tested shift (Δ=2.5; p=0.0026), exceeding the ~½ SD robustness criterion of Ratitch et al. (2013) and confirming strong resilience of the primary endpoint to non-MAR assumptions.

** Primary endpoint at the time of interim analysis (assessed as the average of Day 77 and Day 84)

Source: Praxis October 2025 Prospectus

This appears to us to be a blatant misrepresentation of the very paper that Praxis leans on to support its position. As shown below, the 2013 Ratitch paper relies on an earlier paper by Norman et al., in which the authors argue that a half standard deviation ought to be considered the *"**minimally** important difference"* for health-related quality of

life instruments, citing a wide variety of chronic diseases. The Ratitch paper never appears to describe – as claimed by CEO Souza – that tipping point analysis ought to *end* at 0.5 standard deviations; rather it *begins* there.

Indeed, Ratitch's ***own tipping point analysis provides data at 0.5 standard deviations*** (two HAM-D-17 points, in its case), ***0.75 standard deviations, and 1.0 standard deviations, as shown***:

Table II. Results from analyses using standard multiple imputation (MI) and pattern mixture model (PMM)-based approaches. Change from baseline to week 8 in Hamilton Depression Scale (HAM-D-17) total score.

Analysis	LS mean difference of experimental versus placebo (CIs)	*p*-value
Standard MI (MAR-based)	−4.096 (−6.143, −2.050)	0.0001
Delta-adjusting PMM: $\delta = 2$	−2.995 (−5.157, −0.834)	0.0066
Delta-adjusting PMM: $\delta = 3$	−2.504 (−4.713, −0.295)	0.0263
Delta-adjusting PMM: $\delta = 4$	−2.009 (−4.273, 0.256)	0.0821
Control-based PMM	−3.196 (−5.277, −1.114)	0.0026
PMM based on reasons for discontinuation	−2.395 (−4.602, −0.187)	0.0335

MAR, missing at random; CI, confidence interval; LS, least squares.

Source: Ratich et al.

As shown, in Ratitch's case, the 4-point penalty at 1.0 standard deviation led to a p-value of 0.0821, which is why the author warns that, if such scores are *"plausible clinically for early withdrawers in the experimental arm, then this tipping point analysis would suggest that the MAR result of the study will need to be treated with caution."*[6]

So why does Souza insist on this blatant misreading of the paper? **We believe that the trial result simply would not have survived a tipping point analysis at 0.75 and 1.0 standard deviations.** As said plainly by a former 20-year FDA official we spoke with,

> *"When the company does **really low sensitivity to try to convince the FDA**. To me that's a specific concern."*

By our math, results go up in smoke when the Company simply employs its own cited methodology properly.

- At 0.75x standard deviations, we estimate ulixa shows a 2.8 point decline vs. a Company-favorable assumption of a 1.5 point decline for placebo, for a 1.3-point treatment effect. This remains a statistically significant difference in outcomes, but isn't clinically meaningful according to the Company's own standards by which a patient response is stipulated by a *minimum* decline of at least 3.0 points.[7]

- At a full standard deviation, we estimate ulixa shows a 2.3 point decline vs. a 1.5 point decline for placebo, which is both well below the 3.0 point minimum, and fails to meet statstical significance. [8]

[6] In the case of Ratitch, the endpoint was HAM-D-17 scores, and the standard deviation was 4.
[7] See Study 2, "Patients who improved by 3 points in the mADL11 from baseline were then randomized to receive either placebo or to continue receiving ulixacaltamide" as well as Essential1's disclosure that "Corresponding responder analyses were conducted based on clinically meaningful within-patient change in mADL11 defined as a minimum 3-point improvement."
[8] Given Company-disclosed ½ Std. Dev. difference of 1.7 mADL11 points and p=0.0026, implied Z-score = 3.011454 and SE=0.56451. From here, we can further calculate Z-scores and p-values at ¾ and 1.0x standard deviations.

Tipping Point	mADL11 Shift	Ulixa	Placebo	Difference	p-value
Reported	0.00	(4.3)	(1.7)	(2.6)	<0.0001
1/2 Std. Dev	2.50	(3.3)	(1.6)	(1.7)	0.0026
3/4 Std. Dev	3.75	**(2.8)**	(1.6)	(1.3)	0.0268
1 Std. Dev	5.00	**(2.3)**	(1.5)	(0.8)	**0.1560**

Source: Culper Estimates

By way of an additional sanity check, consider that Praxis' mere 2.5-point shift implies that imputed patients still outperform placebo patients, to say nothing of the possibility that ulixa patients actually perform *worse* than placebo, whether or not a result of ulixa's side effect profile.

Praxis' Butchered Interpretation of Ratitch Further Confounded by High Dropout Rates, Which Requires Even More Penalizing Assumptions

Moreover, Praxis in our view relies on a wholly butchered interpretation of Ratitch, rendered useless by the trial's ridiculously high 35.6% dropout rate. Ratitch and Norman draw primarily on other studies with dropout rates generally in the low single digits, such as asthma and COPD studies. The idea was never to legitimize the wholesale rebuild of a trial lacking massive amounts of completion data. According to the FDA itself, the higher the dropout rate, the harsher the sponsor ought to be in its tipping point analysis.

> The applicant conducted tipping point analyses to evaluate the sensitivity of results obtained from imputation of missing data. The tipping point analyses first imputed missing data using Markov chain Monte Carlo (MCMC) simulations to establish monotonicity of missingness, with simulations based on regression for the timepoint of interest. Data at other timepoints and treatment were used as covariates in these imputations based on the missing-at random assumption. The potential impact of missing data on the de facto estimand was then examined by adding or subtracting 'shifts' to the imputed data. In general, shifts applied by the applicant to the treatment arm were extensively explored, while shifts applied to control arms were minimal. However, because the impact of a shift on mean treatment effect is proportional to the amount of missing data in the trial arm to which the shift is applied, application of only minimal shifts to control arms, or more generally, to trial arms with more trial discontinuations, could omit shifts which lead to reversal of statistically significant results, and which are therefore important to include in the analyses. Therefore, I conducted additional tipping point analysis, with methods described in the following paragraphs.

Source: Olumiant Statistical Review, 2018

Two former FDA officials we spoke told us that the sheer mass of imputed data would make such a tipping point analysis a problem entirely – calling back to earlier concerns that other former FDA officials told us regarding the validity of the entire trial itself.

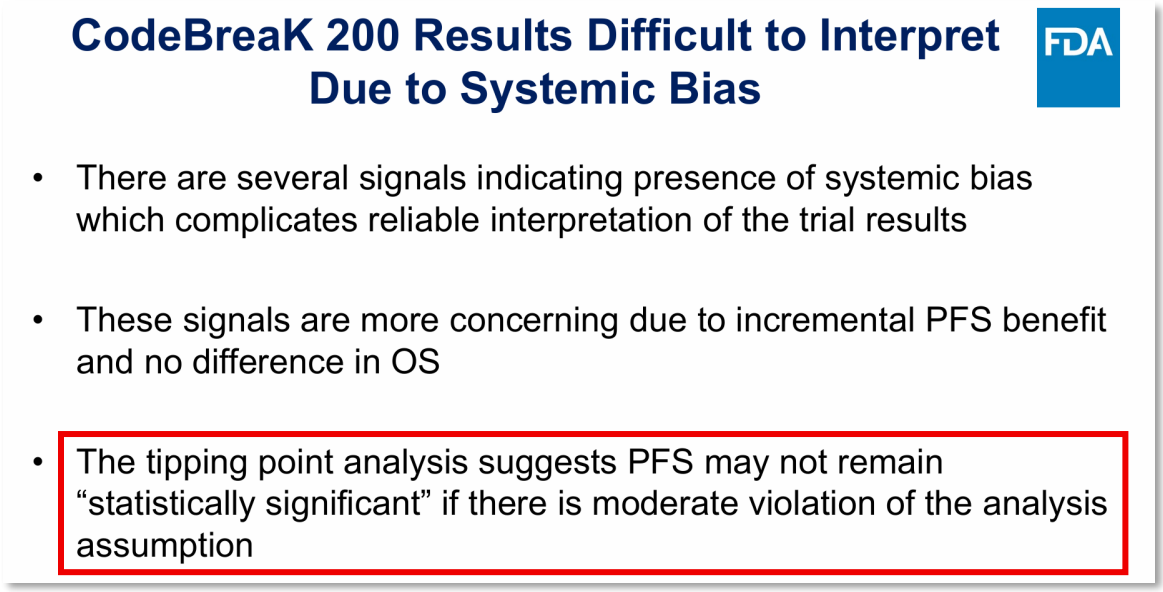
> *"We're not talking about 3% or 5% of missing data, we're talking about a hell of a lot of missing data… If they're hiding behind this imputed data, it'll have to see the light of day. Why did so many drop out, and then number two – the statistical analysis… Yeah, we've got a problem. Houston, we have a problem. I'm leaning towards non-approval until they get this straightened out. Would I buy the stock? I don't think so. There's an imputation of too damn much missing data."* – 40-year FDA Pharmacologist

> *"For a trial above 30% dropout, the trial [data] can't be imputed. The dropout is too high – imputation can't solve the dropout issue."* – Former 10+ year FDA Neurology official

On the October 16, 2025, investor call, CEO Souza implicitly acknowledged that the FDA required a tipping point analysis, stating, "*We prespecified and agreed with the agency that we would conduct a tipping point analysis.*"

Key FDA Officials Have Demonstrated History of Rebuking Nonsense

We believe the FDA will scrutinize the Company's gimmicks closely. To that end, newly appointed CDER Director Richard Pazdur was previously a member of the FDA review team that in October 2023 shot down Amgen's sotorasib, calling confirmatory data "unreliable" and explicitly calling out Amgen's lackluster tipping point analysis in its comments. This rhymes with what we think the FDA will find at Praxis.



Source: Sotorasib AdCom Slides

In May 2024, influential FDA actor Vinay Prasad authored a paper agreeing with Pazdur's conclusion on sotorasib, acknowledging the FDA relied on expanding the tipping point analysis:

> The FDA conducted two sensitivity analyses to show that the progression-free survival benefit of sotorasib over docetaxel could be overestimated under non-extreme assumptions. First, in a tipping point analysis, the FDA analysts made different assumptions regarding the patients in the docetaxel group who dropped out or had early crossover. They found that the HR for the primary endpoint would no longer be significant if these patients had a 50% lower chance of presenting a progression-free survival eventthan those who remained in the docetaxel group. This assumption was not unlikely. Second, the potential overestimation of the

Source: The Lancet

Prasad similarly co-authored a paper in support of the FDA's 2016 denial of Merck's expanded indication for Vytorin, a cardiovascular drug, citing a weak tipping point analysis.

If the IMPROVE-IT Trial Was Positive, as Reported, Why Did the FDA Denied Expanded Approval for Ezetimibe and Simvastatin? An Explanation of the Tipping Point Analysis

Emma P Deloughery [1], Vinay Prasad [2,✉]

Source: NIH

From our review, nothing in Prasad's paper suggests he would tolerate Praxis' rosy tipping point analysis. On the contrary, **Prasad explicitly indicated that in high discontinuation settings, tipping point analysis ought to be tested under harsher, not friendlier, assumptions**:

> *"That the result of the study is more easily "tipped" with higher—arguably more realistic scenarios—event rates indicates that the conclusion is less robust than as presented by the researchers... With the high rate of discontinuations and the greater cardiovascular event rates in the first year, the data is not robust to alternative assumptions regarding the missing data. This calls into question the significance of the results… further study is needed."*



FDA Commissioner Makary, CDER Director Pazdur, and CBER Director Prasad
Source: @DrMakaryFDA

In conclusion, we again recognize the substantial need for safe and effective ET therapies. We're short Praxis for that exact reason – we believe ulixa is the opposite: an ineffective drug with tolerability issues that is being promoted on the back of manipulated, make-believe, and misrepresented data. We expect the FDA reckoning to be swift.